BANK LOAN PORTFOLIO ANALYSIS

End-to-End Banking Data Analytics Project

Python • PostgreSQL • SQL • Power BI

Prepared by

DIVYANSHU GAUTAM

38.6K

LOAN APPLICATIONS ANALYZED

$435.8M

TOTAL FUNDED AMOUNT

WHY THIS MATTERS

Problem Statement

A bank issues thousands of loans every year — but funding a loan is only half the story.

To lend responsibly, a bank must continuously answer: which loans perform well, which borrowers repay on time, and where is risk concentrated?

Without a structured analysis of the loan portfolio, banks risk poor lending decisions, rising defaults, and reduced profitability.

This project analyzes the full loan portfolio to surface lending performance, repayment behavior, and risk — enabling faster, data-driven credit decisions.

Lending Performance

Volume, funding & trends over time

Repayment Behavior

Good loans vs. bad / charged-off loans

Portfolio Risk

DTI, interest rate & risk-level exposure

Decision-Making

Insights that guide lending strategy

WHAT WE SET OUT TO DO

Business Objectives

Loan Performance Monitoring

Track applications, funded amounts, and repayments over time

Good Loan vs. Bad Loan Analysis

Separate performing loans from defaults and charge-offs

Risk Analysis

Assess exposure via DTI, interest rate, and risk level

Customer Segmentation

Profile borrowers by income, employment & home ownership

Profitability & Repayment Insights

Evaluate ROI and recovery across grades and segments

THE FOUNDATION

Dataset Overview

38,576	33	50	2021
Loan Records	Data Columns	US States Covered	Loan Issue Year

KEY FIELDS IN THE DATASET

Loan Amount
Principal disbursed

Interest Rate
Annual cost of borrowing

Loan Status
Fully Paid / Current / Charged Off

Grade & Sub-Grade
Bank's internal risk rating (A–G)

Purpose
Reason for borrowing

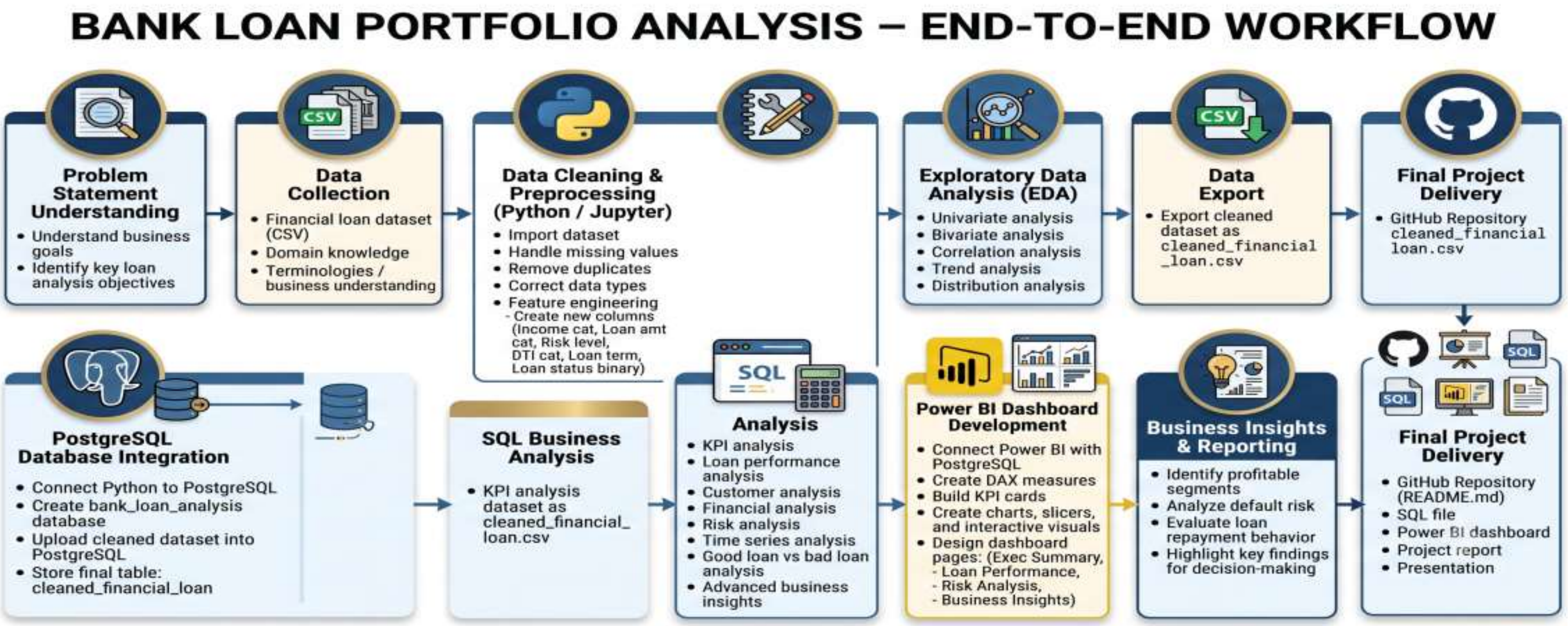
DTI Ratio
Debt-to-income of borrower

Annual Income
Borrower's yearly earnings

Address State
Geographic location of borrower

Project Workflow

CSV Data › Python › PostgreSQL › SQL Analysis › Power BI › Insights › GitHub



Python Data Cleaning & Feature Engineering

DATA CLEANING

- Handled missing & null values across key fields
- Converted issue, payment & credit-pull dates to proper datetime format
- Corrected data types for numeric and categorical fields
- Removed duplicate loan records

Result: a clean, analysis-ready dataset of 38,576 loan records exported for SQL & Power BI.

FEATURE ENGINEERING

Risk Level

Low / Medium / High, derived from DTI & interest rate

Income Category

Low / Middle / Upper-Middle / High income bands

DTI Category

Grouped debt-to-income ranges for segmentation

Loan Term Group

Standardized 36-month vs. 60-month terms

Loan Status Binary

Simplified flag for good loan vs. bad loan

QUERYING THE PORTFOLIO

SQL Business Analysis

74

SQL BUSINESS QUERIES

Written in PostgreSQL against the cleaned_financial_loan table to answer real lending questions.

KPI Analysis

Core lending metrics & portfolio totals

Loan Performance

Trends across time, term & status

Customer Analysis

Borrower profiles & demographics

Risk Analysis

DTI, interest rate & risk exposure

Good vs. Bad Loan

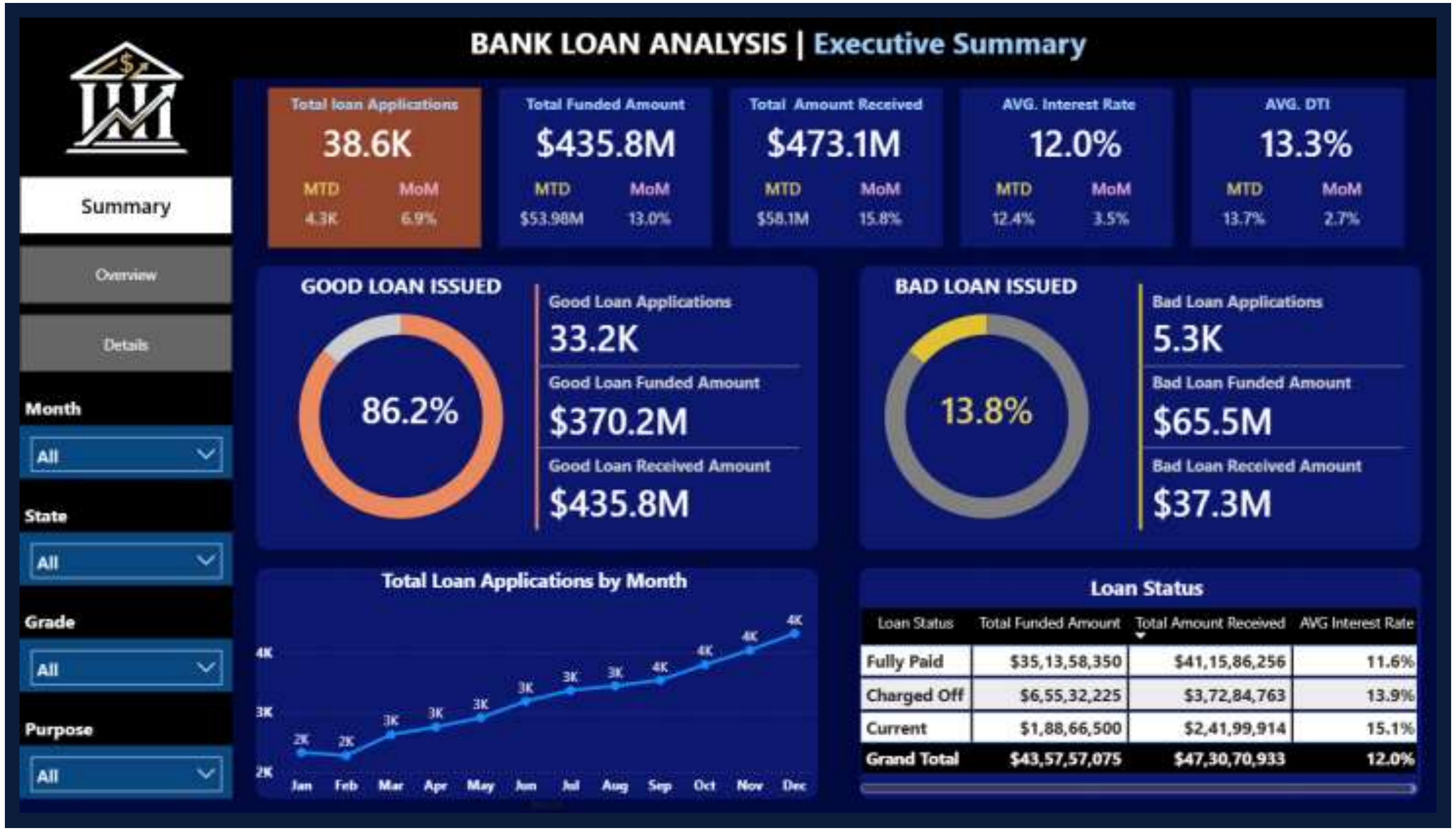
Repayment performance comparison

Advanced Insights

Cross-segment profitability analysis

BRINGING IT TO LIFE

Power BI Dashboard Overview



EXECUTIVE SUMMARY PAGE

OVERVIEW PAGE

DETAILS PAGE

3 interactive pages • KPI cards, slicers, trend charts & maps • Filters by month, state, grade & purpose

WHAT THE DATA TELLS US

Key Business Insights

86.2% Good Loans

33.2K applications classified as Fully Paid or Current

13.8% Default Risk

5.3K loans Charged Off, worth $65.5M in funded amount

Grade A & B Lead

Highest funded volume with lowest historical risk

CA & NY Top States

Highest loan volume and strongest ROI performance

8.56% Portfolio ROI

Return on the $435.8M total funded loan amount

$473.1M Received

Total repayments collected against $435.8M funded

TURNING INSIGHT INTO ACTION

Business Recommendations

Prioritize Profitable, Low-Risk Segments	Grow lending toward Grade A/B borrowers with high repayment consistency
Monitor High-Risk Loan Purposes	Tighten underwriting for purposes with elevated charge-off rates
Strengthen Recovery on Charged-Off Loans	Introduce early-intervention recovery workflows for at-risk accounts
Adopt Grade & Risk-Based Lending	Price interest rates and terms according to borrower risk level
Focus On High-Performing States & Profiles	Expand marketing in top-performing states like CA and NY

Thank You

Questions & Discussion Welcome

GitHub

https://github.com/DivyanshuGautam91

LinkedIn

linkedin.com/in/divyanshu-gautam-a48844298/

Email

gautamdivyanshu47@gmail.com